

Pramod Adhikari, Ph.D.

Research Scientist | Atmospheric Science

Department of Atmospheric Science | University of Wyoming

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EDUCATION

Ph.D. (Atmospheric Science) | University of Nevada, Reno, USA 2018-2022

- Dissertation: “Aerosol-cloud-precipitation interaction based on remote sensing and cloud-resolving modeling over the Central Himalayas”
- Advisor: Dr. John F. Mejia, Associate Research Professor

MS (Atmospheric Science) | University of Nevada, Reno, USA 2015-2017

- MS Thesis: “Assessment of wind regime and sediment transport activity at Oceano Dunes, California”

MSc. (Physics) | Central Department of Physics, Tribhuvan University, Nepal 2011-2015

- MSc. Thesis: “Numerical reconstruction of early monsoon weather situation over Arun valley of eastern Nepal Himalaya”

BSc. (Physics) | St. Xavier’s College, Tribhuvan University, Nepal 2007-2010

KEY SKILLS

Earth System Modeling

WRF & WRF-Chem

Expertise in configuring, modifying, and executing physics-based dynamical downscaling using WRF and WRF-Chem simulations for regional/local level at fine spatial (~1km) and temporal scales (~5 minutes to climate scale).

Climate Data Analysis

Observational (station and satellite), Reanalysis, Global Earth System models (CMIP6, LENS2)

Proficient in collecting, processing, and analyzing regional and global Earth system model large datasets, with a focus on hydroclimate extremes and warming impacts across mountainous terrains

Gridded Data Processing (NetCDF, HDF, GRIB, tiff)

xarray, numpy, scipy, pandas, matplotlib

Skilled in handling and visualizing large-scale spatial and temporal datasets using Python-based tools to generate diagnostics and visualization

High-Performance Computing

Derecho, Casper (NCAR-Wyoming supercomputer)

Extensive experience with supercomputing environments, including parallel processing, PBS job scripting, and data workflow optimization.

Scientific Programming Languages

Python, R, FORTRAN

Strong programming background for data analysis, modeling, and workflow automation. Daily user of Jupyter Notebooks and well-versed in modular, reproducible code design.

Linux/UNIX Environments

Shell Scripting & System Tools

Proficient in Linux-based computing including shell scripting, software environment management, and system-level troubleshooting for research workflows.

Collaborative Research

Team-Oriented Research

Experienced in managing and contributing to interdisciplinary projects with scalable and reproducible workflows, resulting in peer-reviewed publications.

PUBLICATIONS

1. **Adhikari, P.**, Geerts, B., Rahimi S., Rasmussen, K., Quam, M., Huhn, B.,(2026), Charting Resilience at Rocky Mountain Ski Areas as Winters Shrink. *Nature Partner Journals Climate and Atmospheric science* (under review).
2. **Adhikari, P.**, Geerts, B., Rahimi-Esfarjani, S., Xue, L., Rasmussen, K., Smith, K., Gutmann, E., Schneider, T. L., and Dudhia, J., (2026), A 1.33 km resolution climate reconstruction and climate change outlook over the Rocky Mountain Region. *Journal of Applied Meteorology and Climatology* (under review).
3. Smith K., Geerts, B., **Adhikari, P.**, Day K., Rahimi S., Shuman B., Xue L., and Schneider T. (2026) Evaluation of CONUS404 cold-season precipitation and snowpack over the mountainous western United States. *Journal of Hydrometeorology* <https://doi.org/10.1175/JHM-D-25-0125.1>
4. **Adhikari, P.**, Geerts, B., Rahimi-Esfarjani, S., Shuman, B. N., Smith, K., and Day, K. (2025). Global Warming Induced Changes in Extreme Precipitation in the Western United States: Projections from Dynamically Downscaled CMIP6 GCMs. *Geophysical Research Letters* <https://doi.org/10.1029/2025GL116113>
5. Chang, C., Mejia J. F., Henao J. J., and **Adhikari P.**, (2025) Impacts of Wildfire Smoke on Stratocumulus Clouds and Their Diurnal Cycle Using WRF-Chem Modeling, *Journal of Geophysical Research: Atmospheres* <https://doi.org/10.1029/2024JD042405>
6. **Adhikari, P.**, Geerts, B., Tessendorf S., Xue, L., and Schneider, T. L. (2025). Climatology of cold season supercooled liquid water and glaciogenic cloud seeding potential in the western United States, according to a 4 km resolution climate reconstruction. *Journal of Applied Meteorology and Climatology*. <https://doi.org/10.1175/JAMC-D-24-0246.1>
7. **Adhikari, P.**, Geerts, B., Rahimi-Esfarjani, S., Smith, K., Shuman, B. N., & Schneider, T. L. (2024). Evaluation of the mountain hydroclimate across the western United States in dynamically downscaled climate models. *Journal of Hydrometeorology*, 25(12), 1877-1894. <https://doi.org/10.1175/JHM-D-24-0063.1>
8. **Adhikari, P.**, Mejia, J. F. (2023). Aerosol–precipitation elevation dependence over the central Himalayas using cloud-resolving WRF-Chem numerical modeling. *Atmospheric Chemistry and Physics*, 23, 1019-1042. <https://doi.org/10.5194/acp-23-1019-2023>

9. **Adhikari, P., & Mejia, J. F.** (2022). Impact of transported dust aerosols on precipitation over the Nepal Himalayas using convection-permitting WRF-Chem simulation. *Atmospheric Environment: X*, 100179. <https://doi.org/10.1016/j.aeaoa.2022.100179>
10. **Adhikari, P., Mejia, J. F.** (2021). Influence of aerosols on clouds, precipitation and freezing level height over the foothills of the Himalayas during the Indian summer monsoon. *Climate Dynamics* 57, 395–413. <https://doi.org/10.1007/s00382-021-05710-2>

PROFESSIONAL EXPERIENCE

1. University of Wyoming
Research Scientist | April 2026 - present
Postdoctoral Researcher | January 2023 - March 2026
 - Led end-to-end development **of high-resolution regional Earth system modeling project (70+ years at 1.3 km resolution)** from conception through execution. Prepared model setup, designed simulation and automated workflows, performed parallel computing, and managed large-scale data (~PB). Conducted model evaluation and data analysis and collaborated with multidisciplinary teams to produce high-quality datasets that enabled multiple peer-reviewed publications and stakeholder-focused applications.
 - Utilize high-resolution regional climate models to **dynamically downscale global climate simulations**, enhancing predictive understanding of warming on water availability and human-environment systems in the Interior Western U.S.
 - Conduct retrospective performance assessments of downscaled GCMs to establish reliability for local and regional climate impact studies.
 - Assess and project **probabilities of extreme events** (using Generalized Extreme Value distribution) under various **global warming levels** using ensembles of dynamically downscaled CMIP6 data (25 members at 9 km resolution).
 - Contribute modeling and analyses to decision-support tools, including the **WyACT web Portal (wyadapt.org)** and the contributing author to **Snake River Headwaters Futures Assessment (in preparation)** and Colorado Climate Roadmap (*in preparation*), informing extreme event projection and adaptation planning.
 - **Mentor undergraduate and graduate students**, guiding them in big data analysis, modeling workflows, and research methodologies.
2. **Postdoctoral Researcher**
 New Mexico State University | August 2022- January 2023
 - NASA's Planetary Data System: The Planetary Atmospheres Node at the Department of Astronomy
 - Work as part of a team of scientists developing a planetary data system-equivalent archive of atmospheric modeling output through an Atmospheric Modeling Annex.
3. **Graduate Assistant (Ph.D.)**
Research Assistant:
 Desert Research Institute, Reno | 2018- 2022
 - Handle and analyze long-term and large gridded datasets from satellite and reanalysis product

- Lead the project and apply for the funding and computational resources
- Design model experiments and run cloud-resolving WRF-Chem simulations in a high-performance computing environment
- Prepared various anthropogenic, biogenic, fire emission, and initial/boundary condition datasets to be used for WRF-Chem
- Analyze data from the simulation, compile results and prepare a manuscript for publications
- Analyze the data from various sources, e.g., Satellite (e.g., MODIS, CALIPSO, GPM), stations (weather and meteorological stations, AERONET stations), reanalysis (ERA-Interim, ERA5), etc.

Teaching Assistant

University of Nevada, Reno | 2018-2022

- Teach physics laboratory and recitations for undergraduate students majoring in physical sciences, engineers, and non-physical sciences
 - Design syllabus, deliver lectures, conduct a discussion on physics problems, and run lab assignments
 - Ensure a positive learning environment as reflected in the course evaluation
4. **Graduate Research Assistant (M.S.)**, Desert Research Institute, Reno | 2015-2017
- Compiled, validated, and analyzed multi-year datasets to identify the sand transport events over the Oceano Dunes, California
 - Analyzed long-term wind speed data to infer the sand transport potential

CONFERENCE PROCEEDINGS

1. **Adhikari, P.**, et al., (2026; *Invited talk*) Impact of warming climate on western US snowpack and ski areas based on high-resolution Earth system modeling. *Cross Country Ski Area Association annual conference*, Canmore, Canada, 2026.
2. **Adhikari, P.**, et al., (2025) Long-term hydroclimate modeling at 1.33 km resolution over the Rocky Mountains: development and initial results from RM1.3. *American Geophysical Union Fall meeting*, 2025.
3. **Adhikari, P.**, et al., (2025) Predicting weather extremes and water resources in the climate transition: a focus on the Western United States. *American Meteorological Society Annual Meeting*, 2025.
4. **Adhikari, P.**, et al., (2024) Changes in wet and dry extremes in a dynamically downscaled dataset. , *American Geophysical Union Fall Meeting*, 2024.
5. Geerts, B., Albeke, S., Rahimi, S., Shuman, B., **Adhikari, P.**, Williams D., Knapp, C., Bukovsky, M. (2023) Communicating quantitative climate change information to stakeholders and the public: opportunities and challenges of a regional web portal: a case study for Wyoming. *AGU Fall Meeting*, 2023.
6. **Adhikari, P.**, et al., (2023) Elevation-dependence evaluation of historical bias-corrected dynamically downscaled GCMs across the western U.S. *American Geophysical Union Fall Meeting*, 2023.
7. **Adhikari, P.**, Mejia, J. F. (2022). Impact of dust aerosols on the convective system using cloud-resolving WRF-Chem simulation over the Nepal Himalayas. *American Meteorological Society Annual Meeting*, 2022.

8. **Adhikari, P.**, Mejia, J. F. (2020). Impact of transported dust aerosols on precipitation over the central Himalayas using convection permitting WRF-Chem Simulation, *American Geophysical Union Fall Meeting*, 2020.
9. **Adhikari, P.**, Mejia, J. F. (2019). Influence of aerosols on precipitation and vertical temperature distribution over the foothills of Himalayas during the Indian summer monsoon. *American Geophysical Union Fall Meeting*: San Francisco, CA, December 9-13, 2019.

CERTIFICATIONS

1. *Neural Networks and Deep Learning* | June 2025
 - An online non-credit course authorized by DeepLearning.AI and offered through Coursera
2. *Google Advanced Data Analytics* | March 2024
 - An online non-credit course authorized Google and offered through Coursera

OUTREACH

1. Journal Reviewer: Urban Science- MDPI, Atmospheric Research – Science Direct (Elsevier), Climate Dynamics (Springer).
2. Participated in the AGU Science Policy Workshop and Congressional Visit Day (GeoCVD-2025) for sustained federal support of geoscience research and science funding.
3. The Kathmandu Post, **Kathmandu, Nepal**, May 10th, 2021: Air pollution not only impacts health, but can also trigger floods and landslides, study finds (<https://kathmandupost.com/climate-environment/2021/05/09/air-pollution-not-only-impacts-health-but-can-also-trigger-floods-and-landslides-study-finds>)
4. Air Quality News, **United Kingdom**, May 10th, 2021: Air pollution could trigger flooding, study suggests (<https://airqualitynews.com/2021/05/10/air-pollution-could-trigger-flooding-study-suggests/>)

HONORS AND AWARDS

1. *Colin Warden Memorial Endowment*, **Outstanding graduate student researcher**, Desert Research Institute, Reno, Nevada 2021
2. *Graduate Student Association travel award*, University of Nevada, Reno 2018/2019/2022
3. *Charles Francis Cutts scholarship award*, University of Nevada, Reno 2017-2018
4. *International graduate student scholarship award*, University of Nevada, Reno 2016-2017
5. *M.Sc. Fellowship*, Central Department of Physics, Tribhuvan University, Nepal 2011-2013

TRAINING AND WORKSHOPS

- Introduction to the Community WRF-Hydro Modeling System: Interactive Hands-on Tutorial (February 2022)
- Teaching with Technology Course (January-May 2019)
- NCAR Command Language (NCL) and Visualization and Analysis Platform (VAPOR) workshop, (July 2018)
- NASA ARSET applied remote sensing webinar training on “Overview of the Global Disaster Alert and Coordination System” (February 2017)

Field Measurement

- Measurement of threshold shear stress and transport conditions at White Sand Dunes, New Mexico (March 6-11, 2017)
- Wind energy assessment around Kathmandu Valley, Nepal, using SODAR (April to June 2014)

PROFESSIONAL ASSOCIATIONS

- Member, American Geophysical Union
- Member, American Meteorological Society